Q1.

The table below shows the pH in different parts of the digestive system.

part of digestive system	pH
mouth	6.5 – 7.5
stomach	1.0 – 2.5
small intestine	7.0 – 8.0

(a) Amylase is an enzyme which speeds up the digestion of starch.
Amylase is produced by the salivary glands.

Why does amylase stop working in the acidic conditions in the stomach?

……

……

1 mark

(b) As food passes from the stomach into the small intestine, a different digestive juice is mixed with the food.

(i) Use the information in the table to suggest what type of substance this digestive juice is.

……

1 mark

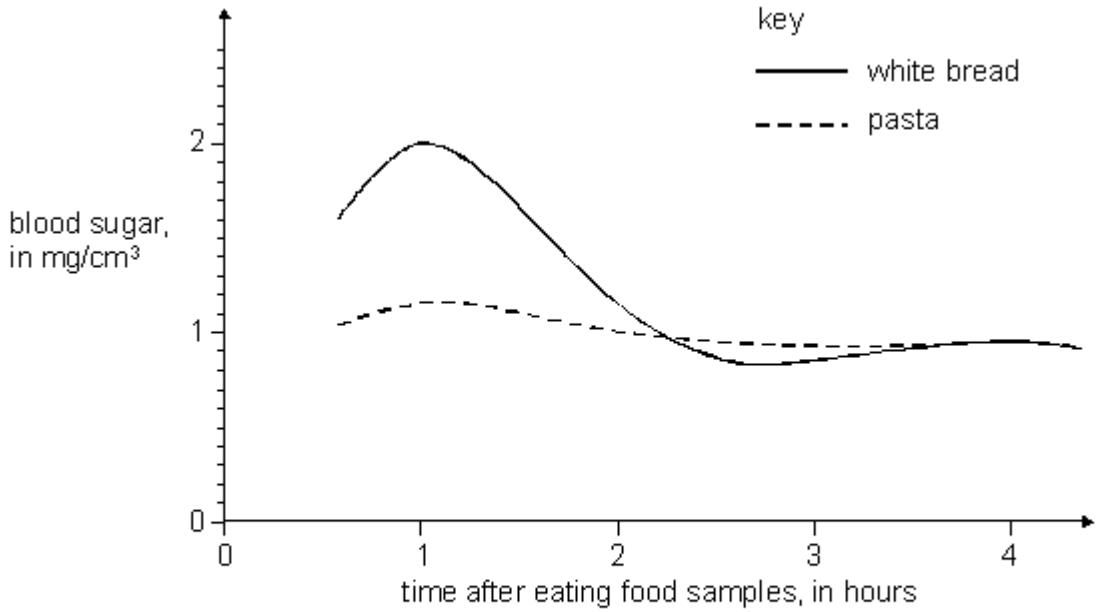
(ii) What type of chemical reaction takes place as this digestive juice mixes with the acid from the stomach?

……

1 mark

(c) After food containing starch is digested, a person's blood sugar level rises. The body responds by producing a chemical called insulin. Insulin lowers the blood sugar level again.

The graph below shows how a person's blood sugar level changes after eating two different samples of food.



(i) Use the graph to help you to explain why less insulin would be produced one hour after eating pasta than one hour after eating white bread.

……

……

1 mark

(ii) Suggest the normal blood sugar level in this case.

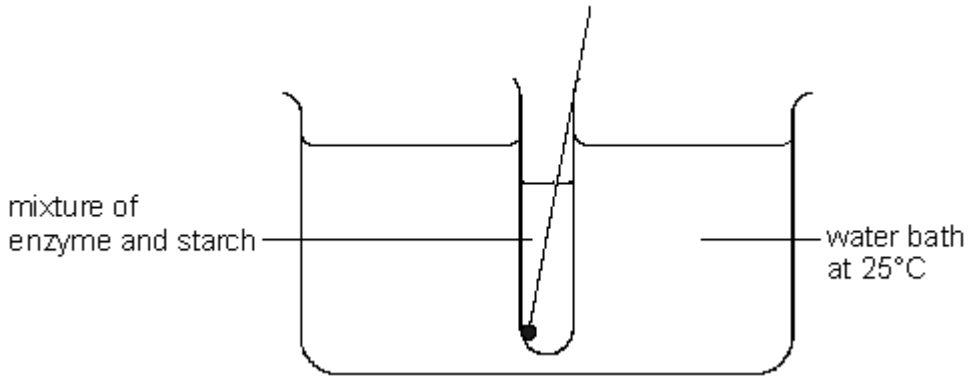
……

1 mark

Maximum 5 marks

Q2.

Sadie and Tom carried out an experiment to investigate the digestion of starch using an enzyme called amylase.



(a) Why was the mixture of enzyme and starch kept in a water bath?

.....

1 mark

Sadie and Tom placed drops of iodine solution on a white tile. They know that starch will turn the iodine solution from brown to dark blue.

Every 30 seconds they added a drop of the mixture of enzyme and starch to a drop of iodine solution on the tile. At first the drops turned blue, but after 240 s they stayed brown.

(b) Why did the mixture stop turning the drops of iodine solution blue after 240 s?

.....

.....

1 mark

(c) They then carried out the experiment with the water bath at 35°C. This time, the drops stopped turning blue after 120 s. How does raising the temperature from 25°C to 35°C affect the digestion of starch?

.....

.....

1 mark

(d) Sadie and Tom want to compare the experiment at 35°C with the results from the experiment at 25°C. Describe what they need to do to make this a fair comparison.

.....

.....

1 mark

(e) A balanced diet includes all of the substances in the following list.

Starch fat fibre minerals protein vitamins

(i) Give the names of the **two** substances in the list which are absorbed into the blood without being digested.

1. .....……

2. .....……

1 mark

(ii) Which substance in the list passes through the body without being digested?

.....……

1 mark

Maximum 6 marks

Q1.

(a) any **one** from

- amylase is denatured **or** destroyed
- enzymes work best at the right pH

accept 'enzymes only work at the right pH'
*do **not** accept 'enzymes do **not** work at this pH'*
*do **not** accept 'amylase does **not** work at this pH'*
*do **not** accept 'amylase is killed at that pH'*

1

(b) (i) an alkali

*accept 'strongly **or** weakly alkaline'*

1

(ii) neutralisation

*do **not** accept 'digestion'*

1

(c) (i) any **one** from

- blood sugar level does not rise as high after eating pasta

accept 'blood sugar is lower'
*do **not** accept 'there is **not** as much starch in pasta'*

- amount of insulin produced is related to the blood sugar level

1

(ii) 0.9 mg/cm^3

accept answers from 0.8 mg/cm^3 to 1.0 mg/cm^3
the unit is required for the mark

1

[5]

Q2.

(a) any **one** from

- to keep the temperature constant

accept 'to keep them at the correct temperature'
*do **not** accept 'to keep them at body temperature'*

- so that the experiment was not affected by a change in temperature in the room
- to control the temperature **or** a variable

1

(b) any **one** from

- the starch had been digested **or** broken down **or** used up

- there was no starch left

1

(c) any **one** from

- it doubled the speed
  *accept 'it made it quicker' **or** 'it increased the speed'*
- it halved the time taken
  accept 'it takes less time'

1

(d) any **one** from

- use the same amount of mixture **or** the same amount of enzyme and starch
  accept 'use the same size drops'
- stir the mixture **or** stir the enzyme and starch
  *accept 'use the same concentration of enzyme **or** starch **or** amylase'*
- use the same type of mixture
  *accept 'use the same concentration of enzyme **or** starch **or** amylase'*

1

(e) (i) minerals and vitamins
answers may be in either order
***both** substances are required for the mark*

1

(ii) fibre

1

[6]